



Plasma proteins associated with disability and mortality risks in Japanese community-dwelling octogenarians

Yusuke Osawa · Takashi Sasaki · Julián Candia ·
Yukiko Abe · Glenn Wong · Stefania Bandinelli ·
Luigi Ferrucci · Yasumichi Arai

Received: 10 April 2026 / Accepted: 15 June 2026
© The Author(s) 2026

Abstract Japan has one of the world's longest life expectancies, yet biomarkers associated with disability and mortality in very old adults remain uncertain. The goal of this study was to identify plasma proteins associated with incident disability and mortality in community-dwelling octogenarians. Two prospective cohorts were analyzed: the Kawasaki Aging Well-being Project (KAWP; 2017–2018; 4.5-year follow-up) as discovery and Invecchiare in Chianti (InCHIANTI; 1998–2000; 15-year follow-up) for external replication. The discovery sample comprised 230 disability-free adults aged 85–89 years (50.4% women). Twenty-nine plasma proteins were assayed

using Luminex xMAP. Associations with incident disability and mortality were estimated with elastic net–regularized Cox models and tested in multi-variable Cox models in the full KAWP cohort and in InCHIANTI. Higher beta-2-microglobulin and cystatin C levels were associated with an increased risk of disability (hazard ratio [HR] 1.35, 95% confidence interval [CI] 1.11–1.64; HR 1.42, 95% CI 1.14–1.76). For mortality, epidermal growth factor was inversely associated (HR 0.51, 95% CI 0.27–0.96), while interferon gamma–induced protein 10 was positively associated (HR 1.23, 95% CI 1.06–1.44). Associations with disability were broadly replicated in the full

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11357-026-02377-7>.

Y. Osawa (✉) · Y. Arai
Graduate School of Health Management, Keio University,
Kanagawa, Japan
e-mail: yosawa@keio.jp

Y. Osawa
Sports Medicine Research Center, Keio University,
Kanagawa, Japan

Y. Osawa · J. Candia · L. Ferrucci
Translational Gerontology Branch, National Institute On
Aging, Baltimore, MD, USA

T. Sasaki · Y. Abe · Y. Arai (✉)
Center for Supercentenarian Medical Research, Keio
University School of Medicine, Tokyo, Japan
e-mail: yasumich@keio.jp

G. Wong
Department of Biological Sciences, National University
of Singapore, Singapore, Singapore

S. Bandinelli
ASL Azienda Sanitaria Locale Toscana Centro, Florence,
Italy

Y. Arai
Faculty of Nursing and Medical Care, Keio University,
Kanagawa, Japan

KAWP cohort and among InCHIANTI participants aged ≥ 80 years, whereas mortality-related findings were not replicated. In conclusion, beta-2-microglobulin and cystatin C may serve as biomarkers of future disability. These patterns implicate kidney dysfunction and inflammation as drivers of health decline in very old adults and support early risk stratification to guide preventive interventions.

Keywords Plasma protein · Disability · Mortality · Long-Term Care · Octogenarians

Introduction

Japan has one of the highest life expectancy rates in the world; people born in 2024 can expect to live up to 87.14 years (women) and 81.09 years (men). Age-related disability is associated with health issues such as social isolation, cognitive decline, and early death [1–3]. Japan established the Long-Term Care Insurance (LTCI) system in 2000 to assist older adults and individuals whose activities of daily living declined due to dementia, cerebrovascular diseases, senility, fractures, and falls [4]. In 2023, 59.5% of individuals aged ≥ 85 years received the LTCI service. As a result, identifying early disability markers is critical for providing preventive services [5] that reduce disability burden and sustain long-term care systems in aging societies.

Proteomic analysis is already used in clinical settings to diagnose diseases and monitor treatment responses. It could help identify biomarkers for poor health outcomes. The Targeting Aging with Metformin workgroup identified potential biomarkers to track age-related changes in health and functional status [6]. Similarly, we previously used a targeted approach to show that plasma biomarkers, such as N-terminal pro-B-type natriuretic peptide (NT-proBNP) and cystatin C, predicted mortality risk in very old populations, including centenarians [7]. While our method validated previous study findings, it was limited to a set of readily available single biomarkers, limiting our knowledge of the biological pathways involved in age-related disability. Large-scale plasma protein scanning, alongside machine learning, could deliver a holistic, individualized health status assessment across multiple health conditions as shown in a previous extensive proteomic study [8].

To advance the current understanding of the mechanisms underlying the development of disability in later life, observational studies using a hypothesis-free, data-driven discovery approach are required. Thus, the purpose of this study was to identify plasma proteins associated with the risk of disability and death in community-dwelling older adults aged 85 and older.

Article types

An observational study in Epidemiology.

Methods

Study population and design

The Kawasaki Aging Well-being Project (KAWP), carried out in Kanagawa, Japan, examines the overall health status and associations with disability in disability-free community-dwelling older adults aged 85–89 years [9]. During the first phase of the KAWP, 1,026 participants went to local facilities in Kawasaki City from March 2017 to December 2018. For the discovery phase of the present analysis, participants who had never needed to be enrolled in the Long-Term Care system (Care level 1 or higher) at the time of study recruitment were chosen at random. Among these participants, approximately one-fifth with complete data on potential covariates (see [Other Variables](#)) were randomly selected for plasma protein measurements owing to the resource and cost constraints inherent to multiplexed proteomic assays. We used machine learning-based variable selection; thus, conventional sample size calculations were not performed owing to the lack of prior knowledge on effect sizes, consistent with previous studies [8]. However, for reference, we estimated the required sample size assuming a conventional Cox model with cystatin C. Based on a previous study [7], assuming alpha of 0.05, beta of 0.10, and event rates of 20–30%, the required sample sizes ranged from 86 to 129 participants (R package, powerSurvEpi v.0.1.3). All participants received an extensive explanation of the study and participated after providing written informed consent. The study protocol conformed to the principles of the Declaration of Helsinki, with the approval of the Ethics Committee of Keio University School of Medicine (ID: 20,160,297). The KAWP study was also

registered in the University Hospital Medical Information Network Clinical Trial Registry as an observational study (ID: UMIN000026053).

Plasma proteins

A total of 38 plasma proteins were quantified on samples obtained from the discovery cohort using the Luminex® xMAP® technology (Supplementary Table S1). The samples were incubated for an entire night in plates at 4°C, with fluorescent-coded magnetic beads, each conjugated with antibodies against the analyte of interest. The plates were then washed and biotinylated antibodies were incubated using the complex. Following incubation, the plates were washed, and Streptavidin-PE was added to each well and incubated. The plates were washed and resuspended in sheath fluid for measurement on the Luminex™ FLEXMAP 3D®. Data were collected using xPONENT® 4.0 (Luminex) software and assessed using Bio-Plex Manager™ Software 6.1.1 (Bio-Rad). Sample concentrations were calculated using a 4PL or 5PL algorithm, depending on the MFI generated by each standard.

We exempted nine proteins from the analysis because more than 40% of the samples fell below the limit of detection (LOD). For the remaining proteins, we replaced values less than LOD with half the minimum value.

Time to disability and death

This study used registry data on LTCI system certification and vital status from Kawasaki city and Kanagawa prefecture. Information about the LTCI is described in Supplementary Table S2. Incident disability was defined as the first certification of Care level 1 (*Yo-kaigo 1*) or higher, a classification in the LTCI. Certifications of LTCI system, as well as death, were registered at the registry office corresponding to the municipality where the decedent lived following the occurrence of these events. The LTCI system had a participant follow-up rate (and data availability) for over 5.5 years of 100% in the discovery phase sub-cohort and 98% in the full KAWP.

Other variables

We conducted face-to-face interviews to assess participants' general characteristics, including age, sex,

education level (high school and higher education or not), smoking status (never smoker or not), and alcohol drinking habits (never drinker or not). Body mass index (BMI; kg/m²) was calculated by weighing participants and measuring their heights. We calculated the glomerular filtration rate (eGFR) by measuring serum creatinine using a standard enzymatic method as follows: $\text{eGFR (mL/min/1.73 m}^2\text{)} = 194 \times [\text{serum creatinine (mg/dL)}]^{-1.094} \times [\text{age (years)}]^{-0.287} \times 0.739$ (if female) [10].

Statistical analysis

We compared deceased and censored individuals, as well as disabled and censored individuals, using the Student's t-test for continuous variables and the chi-square test for categorical variables.

Prior to modeling, protein values falling below the LOD were imputed as described in Sect. 3.2, and only participants with complete covariate data were retained for the subsequent analyses. To investigate candidate proteins associated with the risks of disability and mortality, we fitted elastic net-regularized Cox proportional hazards models (R package, eNetXplorer v. 1.1.3) incorporating the 29 plasma proteins and adjusting for age, sex, BMI, education, smoking/drinking status, and eGFR [11]. The model explores the best-fitted model by varying the mixing parameter alpha between 0 (Ridge) and 1 (Lasso) [12]; alpha was increased in 0.1-step increments. For each alpha, the best shrinkage penalty parameter (lambda) was chosen using a cross-validation procedure that optimized the out-of-bag correlation between observed and predicted outcomes. For this cross-validation procedure, we ran 1,000 iterations, randomly selecting 80% of samples as the training set and 20% as the validation set for each iteration. We chose the parameter alpha with the highest average correlation between observed and predicted results. To estimate the significance of model coefficients, eNetXplorer generates null models by randomly permuting sample labels 250 times during each cross-validation run. Proteins found in the feature selection phase (using a lenient significance threshold of $p < 0.10$) were tested using multivariable Cox hazard models, adjusting for the same covariates as in the previous step. A threshold of $p < 0.10$ was applied as a conservative, hypothesis-generating criterion, consistent with prior proteomic discovery studies [8]. This approach yields more interpretable effect estimates

and confirms the strength of the relationships between these proteins and the outcomes outside of the regularization framework. Furthermore, we developed a composite score to evaluate the combined effect of selected proteins on the risk of outcomes. The Z-scores of each protein were added together, with the signs adjusted to reflect their respective directions of association with the results.

To externally examine our results, we used the data from the Invecchiare in Chianti, “Aging in the Chianti Area” (InCHIANTI) [13]. A total of 954 participants had plasma proteomics data determined by the 1.3 K SomaScan assay (SomaLogic, Boulder, CO) [14–16], disability assessed by Katz ADL scale and subjective 400 m walk capacity [17, 18], and vital status acquired from the Mortality General Registry data operated by the Tuscany Region and death certificates deposited after death at the Registry Office in the municipality of residence. We used the Cox hazards model to investigate the relationships between the selected proteins in the discovery phase and disability and mortality risks after controlling for the same set of covariates. We also conducted the same analysis on a subset of participants aged 80 years and older. Of note, the InCHIANTI cohort differed substantially from the KAWP discovery sample with respect to age distribution (mean age, 66.1 vs. 87.1 years), follow-up duration (15 vs. 4.5 years), disability assessment methodology (Katz Activities of Daily Living and 400-m walk capacity vs. LTCI certification), and proteomic platform (SomaScan 1.3 K vs. Luminex xMAP).

During the external replication phase, we broadened the analysis to the full KAWP, using plasma B2M, cystatin C, NT-proBNP, and fetuin-A measured with orthogonal experimental techniques and their association with disability risk. To assess the technical reproducibility of measurements, we calculated Spearman’s correlations for each of the four proteins, comparing Luminex measurements to those obtained using these orthogonal methods.

The proportional hazards assumption was verified for all Cox models using Schoenfeld residual tests (`cox.zph` in R). In both the KAWP discovery and full-cohort analyses, the global tests were non-significant (all $p > 0.18$). In the InCHIANTI mortality analyses, mild violations of the proportional hazards assumption were observed for the global test of B2M ($p = 0.009$), cystatin C ($p = 0.014$), and epidermal growth factor (EGF; $p = 0.010$). Inspection of the individual Schoenfeld residual terms indicated that these violations

were driven by age ($p < 10^{-8}$) rather than by the protein exposure variables (all protein-specific $p > 0.09$). No violations were observed in any disability analysis (Supplementary Table S3).

P -values below 0.05 were deemed statistically significant. All statistical analyses were carried out using R v.4.3.1.

Ethical considerations

All participants gave written informed consent. The study protocol followed the principles of the Declaration of Helsinki with the authorization of the Ethics Committee of relevant institutional Ethics Committee (ID: 20,160,297).

Results

Participant selection and baseline characteristics

We randomly selected approximately one-fifth ($n = 231$) of the KAWP participants for plasma protein measurement in the discovery cohort. One participant was excluded due to prior certification for LTCI (Support level 1) at enrollment, leaving 230 participants in the analysis of the discovery subcohort (Supplementary Fig. S1), which did not differ from the full KAWP by baseline characteristics (Supplementary Table S4). The mean age of participants was 87.1 years (SD 1.4), and half were women (50%). Among those who died, 26 participants (61.9%) developed disability before death. No baseline characteristics distinguished participants with and without disabilities. Approximately one-fifth of the participants (18.2%) died during the follow-up. Participants who died had similar baseline characteristics to those who did not (Supplementary Table S4).

Relationship between plasma proteins and disability risk

In the feature selection stage, we screened 29 plasma proteins and covariates for associations with disability risk. The best model performance was provided by the elastic net mixing parameter $\alpha = 0.1$ (Fig. 1A). This model returned a concordance index of 0.54. The estimated statistical significance of this model was low ($p = 0.13$). This model yielded a concordance index of 0.54, indicating modest discriminative performance,

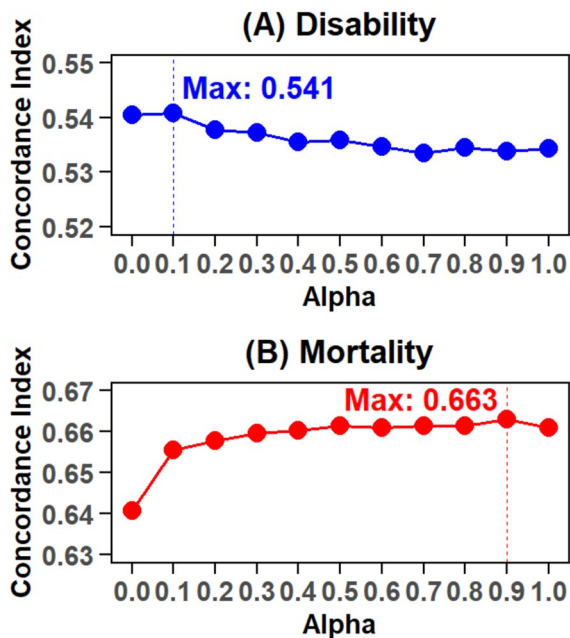


Fig. 1 Cross-validated elastic net Cox hazards model performance assessed using the out-of-bag concordance index as a function of the mixing parameter alpha tuned between 0 (ridge regression) and 1 (lasso) for two outcomes: **A** disability and **B** mortality. For each outcome, alpha was chosen to optimize performance, as shown by the vertical dashed lines

and did not reach statistical significance ($p=0.13$). These findings indicate that the elastic net model was more useful for penalized variable selection than for risk prediction. Nevertheless, B2M and cystatin C were identified as candidate proteins associated with disability risk ($p<0.05$; Fig. 2). By lowering the significance threshold to $p<0.10$, fetuin-A and NT-proBNP were identified as candidate proteins. In the follow-up analysis stage, multivariable Cox models were developed for each one of the selected proteins in model 1 and all selected proteins in one model in model 2 (Table 1). Higher B2M and cystatin C levels were linked to increased disability risk (model 1), while higher cystatin C and lower fetuin-A levels were independently associated with increased disability risk (model 2). The composite score was linked to disability risk (model 3).

Relationship between plasma proteins and mortality risk

The alpha=0.9 elastic net model performed the best (Fig. 1B). The concordance index was 0.66 and the

model was found to be significant ($p<0.01$). EGF was the most predictive plasma protein for mortality risk ($p<0.05$; Fig. 3). By lowering the significance threshold to $p<0.10$, interferon gamma-induced protein 10 (IP-10) and insulin were added as candidate proteins. We built Cox models to complete the feature selection process; EGF, IP-10, and insulin were associated with mortality risk in model 1 (Table 1). The composite score had no association with mortality risk.

External replication using data from the InCHIANTI study

The mean age of participants was 66.1 years (SD 15.2), with 11.5% aged ≥ 80 years. Women comprised 45.8% of the participants. During the 15-year follow-up, 38.2% died (Supplementary Table S5), and higher B2M and cystatin C levels were associated with an increased disability risk. The composite scores were linked to disability risk in all participants and the subgroup of patients aged ≥ 80 years. Mortality-related associations for EGF and IP-10 were not statistically significant in either the overall InCHIANTI cohort or the subgroup aged ≥ 80 years (Table 2).

Consistency of findings in the full KAWP sample

In the entire KAWP, we discovered strong associations between Luminex® xMAP® and other methods ($\rho=0.70$ to 0.89 ; Supplementary Fig. S1). Higher B2M, cystatin C, and NT-proBNP levels were associated with an increased risk of disability when each protein was examined individually. In the model incorporating all four proteins, B2M and NT-proBNP remained independently correlated with disability risk. The composite score was related to disability risk (Supplementary Table S6).

Discussion

We observed that higher risks of disability were associated with elevated levels of B2M and cystatin C, while lower levels of fetuin-A demonstrated a suggestive inverse relationship with disability risk. First, B2M is essential for immune response, and as people age, their levels rise due to various pathological conditions [19]. Accumulated B2M triggers the release of pro-inflammatory cytokines and immune

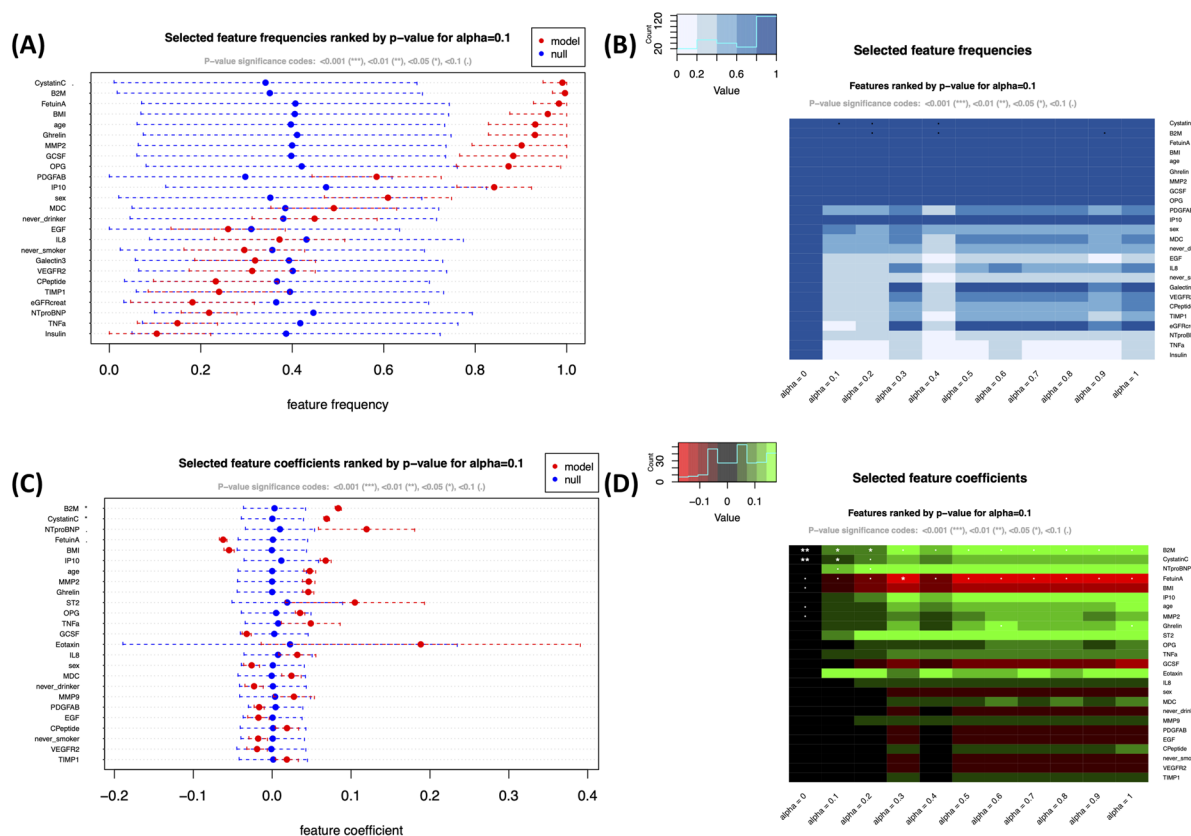


Fig. 2 Feature selection based on elastic net Cox hazards models identifying proteins associated with disability risk (Care 1 certification). **A** Caterpillar plot of feature frequencies sorted by p -value in $\alpha=0.0$. Red symbols (bars) represent the mean (standard deviation) of model feature frequencies across runs. Blue symbols (bars) imply the mean (standard deviation)

cells, which leads to chronic, low-grade inflammation known as “inflammaging” [20]. Elevated B2M has been linked to a higher risk of serious age-related illnesses, including neurodegenerative diseases, cardiovascular diseases, osteoarthritis, and frailty [21]. Second, aging causes elevated cystatin C in the blood. Cystatin C inhibits cysteine proteases, such as cathepsins, which release inflammatory mediators, and controls the activity of immune cells that reduce inflammatory responses [22–24]. Higher cystatin C levels are linked to poor physical and cognitive functions, cardiovascular disease, all-cause mortality, and cause-specific mortality [25, 26]. Although the mechanisms underlying the development of disability in patients with kidney dysfunction are not fully understood, suboptimal kidney function may indicate the severity of subclinical vascular diseases and

of null model feature frequencies across run/permutation null model combinations. **B** The heat map in feature frequencies is sorted by p -value. **C** Caterpillar plot of feature coefficients sorted by p -value in $\alpha=0.0$. **D** The heat map in feature coefficients is sorted by p -value

an accumulation of cardiovascular risk factors [27]. Third, fetuin-A not only prevents ectopic calcification and maintains mineral balance, but it also regulates insulin sensitivity by inhibiting insulin receptor signaling, which is linked to metabolic syndrome and the progression of insulin resistance [28–30]. Metabolic disorders are a major contributor to the development of diabetes-related complications and disability. Building on previous literature, our findings link these proteins to a range of age-related diseases and collectively implicate inflammaging, kidney dysfunction, and multisystem physiological dysregulation as potential biological pathways underlying disability risk in the oldest-old.

In the KAWP discovery cohort, EGF, IP-10, and insulin were associated with mortality risk; however, these associations were not replicated in the external

Table 1 Multivariable Cox hazards models for the associations of selected proteins with the risks of disability and death

	Model 1					Model 2				Model 3					
Protein	HR	95% CI		<i>p</i> -value	C-index	HR	95% CI		<i>p</i> -value	C-index	HR	95% CI		<i>p</i> -value	C-index
Outcome: Disability															
B2M	1.35	1.11	1.64	0.002	0.65	1.26	0.99	1.61	0.056	0.64					
Cystatin C	1.42	1.14	1.76	0.001	0.64	1.33	1.02	1.74	0.032						
Fetuin- A	0.86	0.67	1.10	0.233	0.62	0.77	0.60	0.99	0.041						
NT-proBNP	1.06	0.90	1.24	0.496	0.63	1.00	0.84	1.19	0.989						
Composite score											1.13	1.05	1.23	0.001	0.66
Outcome: Death															
EGF	0.51	0.27	0.96	0.036	0.74	0.45	0.24	0.86	0.015	0.75					
IP-10	1.23	1.06	1.44	0.008	0.72	1.22	1.06	1.42	0.010						
Insulin	1.35	0.99	1.84	0.060	0.72	1.45	1.06	1.98	0.021						
Composite score											1.18	0.99	1.41	0.070	0.73

All models were adjusted for age, sex, education level (high school or not), body mass index, smoking and drinking statuses, and estimated glomerular filtration rate

Model 1 included one protein at a time, while Model 2 included all proteins simultaneously

Of note, all proteins were z-scored. Composite scores were calculated by summing the z-scores of selected proteins after aligning their signs with the direction of the outcome

InCHIANTI replication cohort. Each of these proteins has plausible biological connections to age-related mortality. EGF, primarily involved in cell growth and tissue repair [31], plays context-dependent roles in inflammation and has been linked to chronological aging and age-related diseases [32, 33]. IP-10 contributes to chronic inflammation and immune dysfunction, with elevated levels associated with cardiovascular disease and cancer [34]. Insulin, beyond its central role in glucose metabolism, has been linked to cardiovascular- and cancer-specific mortality in pooled analyses [34, 35]. Nevertheless, because these associations were not reproduced in InCHIANTI, we regard them as hypothesis-generating findings that require confirmation in independent cohorts before their biological relevance in very old adults can be established.

The circulation levels of B2M, cystatin C, and fetuin-A are influenced by kidney function. B2M passes through the glomeruli before being reabsorbed and metabolized in the kidney's proximal tubules [36]. Cystatin C is filtered by the kidney glomerulus and metabolized by the proximal tubules; elevated cystatin C indicates impaired kidney function [6]. Fetuin-A, in addition to its role in mineral metabolism, plays a protective role by binding to calcium and phosphate ions, and preventing their deposition in the renal parenchyma [37]. In this work, 45.9% of the present participants had eGFR values below the threshold for normal

kidney function [38]. This finding suggests that, even at the advanced age of over 85 years, a substantial portion of the participants maintained relatively healthy kidney function. However, kidney function is not independent of the broader context of aging and inflammation. For example, pro-inflammatory cytokines such as C-reactive protein can inhibit fetuin-A synthesis during chronic inflammation; BMI, uric acid, diabetes mellitus, and hypertension all contributed to the link between fetuin-A and decreased kidney function [39]. As a result, the levels of these biomarkers may collectively reflect the complex interaction between inflammatory processes and age-related changes in kidney function. Future research should investigate the temporal sequence of these biomarkers and their associations with kidney function, and then disability risk. It is important to acknowledge that B2M, cystatin C, NT-proBNP, and EGF are substantially influenced by kidney function, cardiovascular disease burden, and systemic inflammation. Notably, B2M and cystatin C have biological functions beyond glomerular filtration. B2M, a component of the major histocompatibility complex class I molecule, accumulates through processes of inflammaging, while cystatin C serves as an endogenous inhibitor of cysteine proteases and modulates immune cell activity [40, 41]. These proteins may therefore capture inflammatory and immune-aging mechanisms not fully indexed by eGFR alone, offering

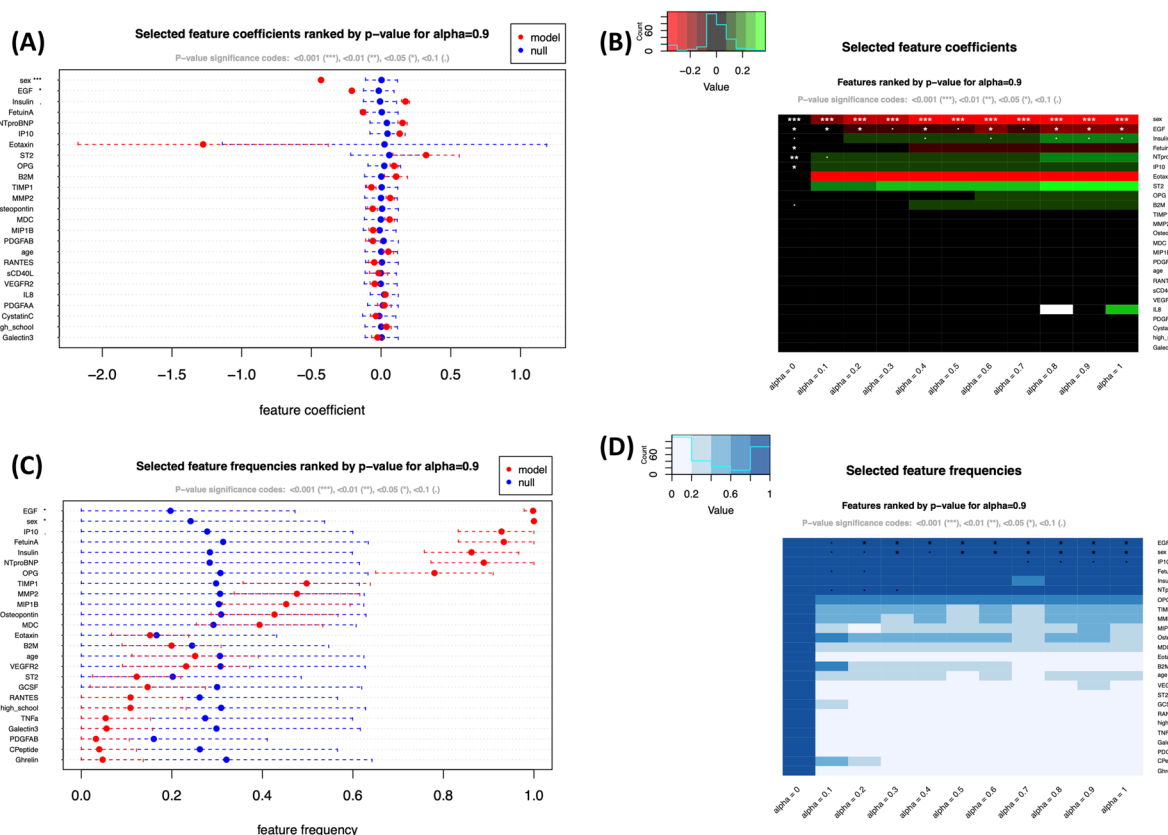


Fig. 3 Feature selection based on elastic net Cox models identifying proteins associated with mortality risk. **A** Caterpillar plot of feature frequencies sorted by p -value in $\alpha=0.9$ Red symbols (bars) implies the mean (standard deviation) of model feature frequencies between runs. Blue symbols (bars) indicate

a plausible biological explanation for the residual associations observed after adjustment for eGFR. Accordingly, the present study cannot determine whether these proteins reflect primary biological aging processes or downstream manifestations of accumulated disease burden and physiological decline. Future studies incorporating direct measures of biological age (e.g., epigenetic clocks), comprehensive comorbidity indices, and longitudinal biomarker assessments will be needed to disentangle these pathways.

While no shared proteins were discovered for risks of disability and mortality, B2M, cystatin C, fetuin-A, and EGF were found to be part of the senescence-associated secretory phenotype (SASP) (<http://www.saspatlas.com/>) [42]. Senescent cells secrete SASPs, a complex mixture of molecules that include cytokines, chemokines, and growth factors, which are involved in both pro- and anti-inflammatory processes. Significant

the mean (standard deviation) of null model feature frequencies across run/permutation null model combinations. **B** The heat map in feature frequencies is sorted by p -value. **C** Caterpillar plot of feature coefficients sorted by p -value in $\alpha=0.9$. **D** The heat map in feature coefficients is sorted by p -value

proteins include both pro- and anti-inflammatory proteins. The balance of pro- and anti-inflammatory components in SASPs may differ depending on physical status, but it could be a risk factor for developing disability later in life. However, the present study measured only a limited panel of circulating proteins and did not directly assess senescent cell burden, immune-aging trajectories, or biological age acceleration. Therefore, the overlap of the identified proteins with the SASP atlas should be regarded as hypothesis-generating rather than mechanistically conclusive. Direct experimental evidence establishing a causal role of SASP factors in the pathogenesis of disability among very old adults remains lacking.

The current study's strength is its longitudinal design, which followed up on community-dwelling octogenarians who are predisposed to developing disability, allowing for the identification of

Table 2 Protein-specific multivariable Cox hazards models for the association between selected plasma proteins and the risks of disability and all-cause mortality in InCHIANTI participants

All models adjusted for age, sex, education, BMI, smoking, and drinking. HR per 1 SD increase in protein (z-scored)

Protein	Overall (<i>n</i> = 954)				Age ≥ 80 (<i>n</i> = 110)			
	HR	95%CI		<i>p</i> -value	HR	95%CI		<i>p</i> -value
Outcome: ADL Disability								
B2M	1.33	1.21	1.46	<0.001	1.65	1.33	2.03	<0.001
Cystatin C	1.36	1.21	1.54	<0.001	1.56	1.24	1.96	<0.001
NT-proBNP	1.19	1.03	1.37	0.019	1.20	1.02	1.42	0.031
EGF	0.94	0.74	1.19	0.582	0.89	0.57	1.39	0.602
IP-10	0.91	0.71	1.16	0.432	0.89	0.42	1.92	0.775
Outcome: Physical Disability								
B2M	1.38	1.27	1.51	<0.001	1.50	1.24	1.81	<0.001
Cystatin C	1.42	1.27	1.59	<0.001	1.41	1.15	1.74	0.001
NT-proBNP	1.21	1.06	1.37	0.004	1.20	1.02	1.40	0.024
EGF	0.88	0.71	1.10	0.278	0.91	0.60	1.37	0.644
IP-10	0.93	0.74	1.16	0.506	1.09	0.55	2.17	0.809
Outcome: Mortality								
B2M	1.16	1.10	1.23	<0.001	1.29	1.13	1.48	<0.001
Cystatin C	1.17	1.10	1.26	<0.001	1.32	1.11	1.56	0.001
NT-proBNP	1.32	1.22	1.42	<0.001	1.15	1.03	1.28	0.014
EGF	0.89	0.78	1.01	0.076	0.75	0.56	1.01	0.058
IP-10	1.06	0.97	1.16	0.164	1.27	0.81	1.97	0.294

disability in a relatively short period. Furthermore, we used a comprehensive database that contained information about certification by the LTCI system and vital status. The consistency of the associations between B2M and cystatin C and disability risk across the full KAWP cohort and the external InCHIANTI cohort further supports the robustness of these findings. In contrast, the associations of EGF and IP-10 with mortality risk observed in the KAWP cohort were not replicated in the InCHIANTI cohort, underscoring the need for cautious interpretation and further replication in independent populations. However, there were some limitations. First, the inclusion of relatively healthy disability-free octogenarians at enrollment, as well as the fact that study participants underwent extensive health checks, including physical performance tests in hospitals, may limit the generalizability of the findings to this age group. Second, data on the reasons for Care level 1 certification were unavailable for all participants because it was not provided for our study; although, the data were recorded by the authorities. Finally, not all participants who developed disability required the LTCI, raising the possibility of disability misclassification.

Conclusions

This study found that in a cohort of community-dwelling octogenarians, plasma B2M, and cystatin C were associated with disability risk. Fetuin-A also showed a suggestive inverse association with disability risk in the KAWP discovery cohort. Furthermore, EGF, IP-10, and insulin levels were associated with mortality risk in the KAWP, although these associations were not replicated in the InCHIANTI cohort. Future studies incorporating longitudinal biomarker measurements are needed to clarify the causal roles of these proteins in the aging process.

Author contribution Yusuke Osawa and Yasumichi Arai designed the study. Yusuke Osawa, Yasumichi Arai, Takashi Sasaki, Yukiko Abe, and Glenn Wong collected the KAWP data. Luigi Ferrucci, Stefania Bandinelli, and Julián Candia collected the InCHIANTI data. Yusuke Osawa, Julián Candia, and Yasumichi Arai analyzed and interpreted the data. Yasumichi Arai and Julián Candia critically revised the manuscript. All authors approved the final version of the manuscript.

Funding This work was supported by JSPS KAKENHI (Grant Number, 21K10505, 24K02711, and 23H03337), Keio Global Research Institute Pre-Start-up Grant (T04EZ00011), Japan Agency for Medical Research and Development

(Grant Number, 17jm0210051h0001, 23zf0127007h0002, 23ae0121041h0003), JST Research Complex Program (JP15667051), and grants from Kanagawa Institute of Industrial Science and Technology. LF and JC were supported entirely by the Intramural Research Program of the National Institute on Aging, NIH (USA). The InCHIANTI study was funded by grants from the National Institutes of Health (R01HL111271 and R21HL112662 to Dr. Ferrucci) and National Institute on Aging Contract 263MD9164 (to Dr. Ferrucci) and was supported by the Intramural Research Program of the National Institute on Aging, National Institutes of Health. The InCHIANTI study baseline (1998–2000) was supported as a “targeted project” (ICS110.1/RF97.71) by the Italian Ministry of Health and in part by the U.S. National Institute on Aging (Contracts: 263 MD 9164 and 263 MD 821336); the InCHIANTI Follow-up 1 (2001–2003) was funded by the U.S. National Institute on Aging (Contracts: N.1-AG-1-1 and N.1-AG-1-2111); and the InCHIANTI Follow-ups 2 and 3 studies (2004–2010) were financed by the U.S. National Institute on Aging (Contract: N01-AG-5-0002) and supported in part by the Intramural Research Program of the National Institute on Aging, National Institutes of Health, Baltimore, Maryland.”

Data availability Data are available upon reasonable request to the study steering committee, subject to approval by the relevant Ethics Committee. InCHIANTI data are available upon request through proposals submitted via the InCHIANTI study website.

Code availability Elastic net models were implemented using the R package eNetXplorer (v.1.1.3), available at <https://github.com/juliancandia/eNetXplorer>.

Declarations

Ethics approval The KAWP study protocol was approved by the Ethics Committee of Keio University School of Medicine (ID: 20160297) and was conducted in accordance with the Declaration of Helsinki. The InCHIANTI study protocol was approved by the Italian National Institute of Research and Care on Aging Ethics Committee and by the Ethics Committee of the Florence ASL 10 regional health authority.

Consent to participate Written informed consent was obtained from all participants.

Consent for publication Not applicable.

Competing interests The authors declare that they have no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated

otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Forman-Hoffman VL, Ault KL, Anderson WL, Weiner JM, Stevens A, Campbell VA, et al. Disability status, mortality, and leading causes of death in the United States community population. *Med Care*. 2015;53(4):346–54. <https://doi.org/10.1097/MLR.0000000000000321>.
- Gomez-Zuniga B, Pousada M, Armayones M. Loneliness and disability: a systematic review of loneliness conceptualization and intervention strategies. *Front Psychol*. 2022;13:1040651. <https://doi.org/10.3389/fpsyg.2022.1040651>.
- Collaborators GBDA. Global, regional, and national burden of diseases and injuries for adults 70 years and older: systematic analysis for the Global Burden of Disease 2019 Study. *BMJ*. 2022;376:e068208. <https://doi.org/10.1136/bmj-2021-068208>.
- Cabinet Office Government of Japan. Annual Report on Aging Society 2021. Tokyo 2021.
- World Health Organization: Health data overview for Japan. 2019. <https://data.who.int/countries/392>. Accessed 8/1 2023.
- Justice JN, Ferrucci L, Newman AB, Aroda VR, Bahnson JL, Divers J, et al. A framework for selection of blood-based biomarkers for geroscience-guided clinical trials: report from the TAME Biomarkers Workgroup. *GeroScience*. 2018;40(5–6):419–36. <https://doi.org/10.1007/s11357-018-0042-y>.
- Hirata T, Arai Y, Yuasa S, Abe Y, Takayama M, Sasaki T, et al. Associations of cardiovascular biomarkers and plasma albumin with exceptional survival to the highest ages. *Nat Commun*. 2020;11(1):3820. <https://doi.org/10.1038/s41467-020-17636-0>.
- Williams SA, Kivimaki M, Langenberg C, Hingorani AD, Casas JP, Bouchard C, et al. Plasma protein patterns as comprehensive indicators of health. *Nat Med*. 2019;25(12):1851–7. <https://doi.org/10.1038/s41591-019-0665-2>.
- Arai Y, Oguma Y, Abe Y, Takayama M, Hara A, Urushihara H, et al. Behavioral changes and hygiene practices of older adults in Japan during the first wave of COVID-19 emergency. *BMC Geriatr*. 2021;21(1):137. <https://doi.org/10.1186/s12877-021-02085-1>.
- Matsuo S, Imai E, Horio M, Yasuda Y, Tomita K, Nitta K, et al. Revised equations for estimated GFR from serum creatinine in Japan. *Am J Kidney Dis*. 2009;53(6):982–92. <https://doi.org/10.1053/j.ajkd.2008.12.034>.
- Candia J, Tsang JS. eNetXplorer: an R package for the quantitative exploration of elastic net families for generalized linear models. *BMC Bioinformatics*. 2019;20(1):189. <https://doi.org/10.1186/s12859-019-2778-5>.
- Zou H, Hastie T. Regularization and variable selection via the elastic net. *J R Stat Soc B (statistical methodology)*. 2005;67(2):301–20.

13. Ferrucci L, Bandinelli S, Benvenuti E, Di Iorio A, Macchi C, Harris TB, et al. Subsystems contributing to the decline in ability to walk: bridging the gap between epidemiology and geriatric practice in the InCHIANTI study. *J Am Geriatr Soc.* 2000;48(12):1618–25. <https://doi.org/10.1111/j.1532-5415.2000.tb03873.x>.
14. Kraemer S, Vaught JD, Bock C, Gold L, Katilius E, Keeney TR, et al. From SOMAmer-based biomarker discovery to diagnostic and clinical applications: a SOMAmer-based, streamlined multiplex proteomic assay. *PLoS ONE.* 2011;6(10):e26332. <https://doi.org/10.1371/journal.pone.0026332>.
15. Candia J, Cheung F, Kotliarov Y, Fantoni G, Sellers B, Griesman T, et al. Assessment of variability in the SOMAscan assay. *Sci Rep.* 2017;7(1):14248. <https://doi.org/10.1038/s41598-017-14755-5>.
16. Candia J, Daya GN, Tanaka T, Ferrucci L, Walker KA. Assessment of variability in the plasma 7k SomaScan proteomics assay. *Sci Rep.* 2022;12(1):17147. <https://doi.org/10.1038/s41598-022-2116-0>.
17. Osawa Y, Semba RD, Fantoni G, Candia J, Biancotto A, Tanaka T, et al. Plasma proteomic signature of the risk of developing mobility disability: a 9-year follow-up. *Aging Cell.* 2020;19(4):e13132. <https://doi.org/10.1111/ace1.13132>.
18. Wallace M, Shelkey M, Hartford Institute for Geriatric N. Katz index of independence in activities of daily living (ADL). *Urol Nurs.* 2007;27(1):93–4.
19. Kang JS, Yang YR. Circulating plasma factors involved in rejuvenation. *Aging (Albany NY).* 2020;12(22):23394–408. <https://doi.org/10.18632/aging.103933>.
20. Smith LK, He Y, Park JS, Bieri G, Snethlage CE, Lin K, et al. beta2-microglobulin is a systemic pro-aging factor that impairs cognitive function and neurogenesis. *Nat Med.* 2015;21(8):932–7. <https://doi.org/10.1038/nm.3898>.
21. Muneshige K, Uchida K, Aikawa J, Iwase D, Takano S, Miyagi M, et al. beta2-microglobulin alters the profiles of inflammatory cytokines and of matrix metalloprotease in macrophages derived from the osteoarthritic synovium. *Cent Eur J Immunol.* 2022;47(4):332–8. <https://doi.org/10.5114/ceji.2022.124092>.
22. Odden MC, Tager IB, Gansevoort RT, Bakker SJ, Katz R, Fried LF, et al. Age and cystatin C in healthy adults: a collaborative study. *Nephrol Dial Transplant.* 2010;25(2):463–9. <https://doi.org/10.1093/ndt/gfp474>.
23. Chattopadhyay M, Mukherjee S, Chatterjee SK, Chattopadhyay D, Das S, Majumdar SS, et al. Impairment of energy sensors, SIRT1 and AMPK, in lipid induced inflamed adipocyte is regulated by Fetuin A. *Cell Signal.* 2018;42:67–76. <https://doi.org/10.1016/j.cellsig.2017.10.005>.
24. Hennige AM, Staiger H, Wicke C, Machicao F, Fritsche A, Haring HU, et al. Fetuin-A induces cytokine expression and suppresses adiponectin production. *PLoS ONE.* 2008;3(3):e1765. <https://doi.org/10.1371/journal.pone.0001765>.
25. Hart A, Blackwell TL, Paudel ML, Taylor BC, Orwoll ES, Cawthon PM, et al. Cystatin C and the Risk of Frailty and Mortality in Older Men. *J Gerontol A Biol Sci Med Sci.* 2017;72(7):965–70. <https://doi.org/10.1093/gerona/glw223>.
26. Onopiuk A, Tokarzewicz A, Gorodkiewicz E. Cystatin C: a kidney function biomarker. *Adv Clin Chem.* 2015;68:57–69. <https://doi.org/10.1016/bs.acc.2014.11.007>.
27. Sarnak MJ, Katz R, Fried LF, Siscovick D, Kestenbaum B, Seliger S, et al. Cystatin C and aging success. *Arch Intern Med.* 2008;168(2):147–53. <https://doi.org/10.1001/archinternmed.2007.40>.
28. Hsu MC, Chen CH, Wang MC, Chen WH, Hu PA, Guo BC, et al. Apigenin targets fetuin-A to ameliorate obesity-induced insulin resistance. *Int J Biol Sci.* 2024;20(5):1563–77. <https://doi.org/10.7150/ijbs.91695>.
29. Yamasandhi PG, Dharmalingam M, Balekuduru A. Fetuin-A in newly detected type 2 diabetes mellitus as a marker of non-alcoholic fatty liver disease. *Indian J Gastroenterol.* 2021;40(6):556–62. <https://doi.org/10.1007/s12664-021-01176-6>.
30. Kim JE, Kim JS, Jo MJ, Cho E, Ahn SY, Kwon YJ, et al. The roles and associated mechanisms of adipokines in development of metabolic syndrome. *Molecules.* 2022. <https://doi.org/10.3390/molecules27020334>.
31. Cortvrindt C, Speckaert R, Delanghe JR, Speckaert MM. Urinary epidermal growth factor: a promising, “next generation” biomarker in kidney disease. *Am J Nephrol.* 2022;53(5):372–87. <https://doi.org/10.1159/000524586>.
32. Cheaito K, Bahmad HF, Jalloul H, Hadadeh O, Msheik H, El-Hajj A, et al. Epidermal growth factor is essential for the maintenance of novel prostate epithelial cells isolated from patient-derived organoids. *Front Cell Dev Biol.* 2020;8:571677. <https://doi.org/10.3389/fcell.2020.571677>.
33. Carlin CR. Role of EGF receptor regulatory networks in the host response to viral infections. *Front Cell Infect Microbiol.* 2021;11:820355. <https://doi.org/10.3389/fcimb.2021.820355>.
34. Ghanavati M, Rahmani J, Rinaldi G, Zand H. Fasting insulin and risk of cancer related mortality in non-diabetic adults: a dose-response meta-analysis of cohort studies. *Curr Diabetes Rev.* 2020;16(4):357–63. <https://doi.org/10.2174/1573399815666190906130544>.
35. Group DIS. Plasma insulin and cardiovascular mortality in non-diabetic European men and women: a meta-analysis of data from eleven prospective studies. *Diabetologia.* 2004;47(7):1245–56. <https://doi.org/10.1007/s00125-004-1433-4>.
36. Vincent C, Chanard J, Caudwell V, Lavaud S, Wong T, Revillard JP. Kinetics of 125I-beta 2-microglobulin turnover in dialyzed patients. *Kidney Int.* 1992;42(6):1434–43. <https://doi.org/10.1038/ki.1992.438>.
37. ChekolAbebe E, TilahunMuche Z, Behaile TMA, MengieAyele T, MekonnenAgidew M, TeshomeAzezew M, et al. The structure, biosynthesis, and biological roles of fetuin-A: a review. *Front Cell Dev Biol.* 2022;10:945287. <https://doi.org/10.3389/fcell.2022.945287>.
38. Sarnak MJ, Levey AS. Cardiovascular disease and chronic renal disease: a new paradigm. *Am J Kidney Dis.* 2000;35(4 Suppl 1):S117–31. [https://doi.org/10.1016/s0272-6386\(00\)70239-3](https://doi.org/10.1016/s0272-6386(00)70239-3).
39. Bassey PE, Numthavaj P, Rattanasiri S, Sritara P, McEvoy M, Ongphiphadhanakul B, et al. Causal association pathways between fetuin-A and kidney function: a mediation analysis. *J Int Med Res.* 2022;50(4):03000605221082874.

40. Zi M, Xu Y. Involvement of cystatin C in immunity and apoptosis. *Immunol Lett.* 2018;196:80–90. <https://doi.org/10.1016/j.imlet.2018.01.006>.
41. Chiou SJ, Ko HJ, Hwang CC, Hong YR. The double-edged sword of beta2-microglobulin in antibacterial properties and amyloid fibril-mediated cytotoxicity. *Int J Mol Sci.* 2021. <https://doi.org/10.3390/ijms22126330>.
42. Basisty N, Kale A, Jeon OH, Kuehnemann C, Payne T, Rao C, et al. A proteomic atlas of senescence-associated secretomes for aging biomarker development. *PLoS Biol.* 2020;18(1):e3000599. <https://doi.org/10.1371/journal.pbio.3000599>.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.